

## **Rhino Hunting**

### **Digital Forensic Investigation Report**

#### ***DFRWS 2005 “Rodeo” Evidence Set USB Image, FTP, HTTP and Executable Traffic Analysis***

**Case ID:** CIP-B104-CS3-C11/26/DFIT/17305

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**Registration Number:** C11/26/DFIT/17305

**Module:** CIP-B104 Case Study Three (3)

**Date of Examination:** 8<sup>th</sup> September, 2026

**Evidence Item(s):** RHINOUSB.dd (FAT16 USB image), rhino.log, rhino2.log, rhino3.log  
(Ethernet/TCP packet captures)

#### **Student Declaration**

I confirm that this report and the analysis presented in it are my own independent work. Every extraction, hash, and finding in this report was produced by executing forensic tools (The Sleuth Kit, tshark, foremost, John the Ripper, exiftool ) directly against my own copy of the assigned evidence in my own Kali Linux environment. I did not copy the conclusions in this report from a walkthrough prepared for a different case.

**Signed:** Nasiru Lawal

**Date:** 8<sup>th</sup> September, 2026

## 2. Executive Summary

In this investigation, I examined a FAT16 USB drive image (RHINOUSB.dd) and three network packet captures (rhino.log, rhino2.log, rhino3.log) associated with the fictional “Rhino Hunting” case. I verified the integrity of all four evidence files using MD5 and SHA1 hashes, and the values remained consistent throughout my examination.

The USB file system contained only two allocated files (both decoy recipe text files) with no deleted or orphaned entries recoverable through standard file-system metadata which is consistent with a quick-format or metadata reset rather than normal file by file deletion. Signature based carving of unallocated space recovered 9 additional JPEG/GIF files, one of which was a byte identical duplicate of another, yielding 8 unique carved images.

My analysis of rhino.log identified three FTP upload sessions from account ‘gnome’ to server ‘cook’, transferring rhino1.jpg, rhino3.jpg (uploaded twice, once corrupted in ASCII mode), and an encrypted archive, contraband.zip. The archive's encryption was confirmed at the byte level and its password was recovered using John the Ripper against the rockyou.txt wordlist (no specific wordlist was named in the assignment brief), yielding the password ‘monkey’.

My analysis of rhino2.log recovered two files (rhino4.jpg, rhino5.gif) downloaded via HTTP from a directory listing at [www.cs.uno.edu/~gnome](http://www.cs.uno.edu/~gnome). My analysis of rhino3.log recovered a Windows executable transferred from the same server, following a Google search for ‘rhino.exe’ and a request to [/~gnome/rhino.exe](http://~gnome/rhino.exe) static analysis identified this file as a legitimate copy of Microsoft's diskpart.exe utility, not an identified malicious payload, based on embedded version-resource metadata.

After combining the results from all sources, I identified 12 visually confirmed unique rhinoceros images across the USB carve set and network captures, meeting the case's fictional threshold of nine or more.

### 3. Authority, Scope and Investigative Questions

#### 3.1 Authority and Engagement

I conducted this examination under the CIP-B104 Module 3 “Rhino Hunting” case assignment, using the DFRWS 2005 forensic challenge evidence set (RHINOUSB.dd, rhino.log, rhino2.log, rhino3.log) supplied through course materials.

#### 3.2 Scope

- In scope: RHINOUSB.dd and the three packet captures, examined entirely offline.
- Out of scope: connecting any recovered torrent, tracker, peer, or server address found in the evidence to a live network; executing the recovered executable outside an instructor-controlled sandbox; any device or account not represented in the four evidence files themselves.
- I performed the examination in an isolated Kali Linux virtual machine with no bridged network access to the evidence's original hosts, trackers, or FTP/HTTP servers.

#### 3.3 Investigative Questions (from the assignment's 8 Required Investigation Work sections)

- 1. Evidence preparation and provenance control was the evidence hashed, logged, and handled with an auditable chain of custody?
- 2. USB image and deleted-file recovery what is the FAT16 layout, and what allocated/deleted/carved files exist?
- 3. Steganography examination is there evidence of hidden payloads in any non-rhino carrier image, and can it be extracted?
- 4. FTP traffic analysis what files were transferred by FTP, by whom, and what does the encrypted archive contain?
- 5. HTTP, optional email, and executable traffic analysis what image and executable objects were transferred by HTTP, and what is the executable?
- 6. Cross-source image correlation and unique count how many distinct, verified rhino images exist across all sources?

- 7. Timeline and final forensic opinion what is the UTC chronology, and what conclusion does the evidence support?
- 8. Sensitive evidence disposal has authorized derived evidence been securely destroyed and logged at the appropriate time?

## 4. Evidence Inventory, Chain of Custody and Integrity Verification

### 4.1 Evidence Inventory

| Evidence ID | Description                       | Size              | Source                                         |
|-------------|-----------------------------------|-------------------|------------------------------------------------|
| EV-01       | RHINOUSB.dd FAT16 USB drive image | 259,506,176 bytes | DFRWS2005 Rodeo evidence set, course materials |
| EV-02       | rhino.log packet capture          | 3,187,907 bytes   | DFRWS2005 Rodeo evidence set                   |
| EV-03       | rhino2.log packet capture         | 292,604 bytes     | DFRWS2005 Rodeo evidence set                   |
| EV-04       | rhino3.log packet capture         | 226,094 bytes     | DFRWS2005 Rodeo evidence set                   |

### 4.2 Integrity Verification (Hash Manifest Original Evidence)

Hashes were recorded at the start of examination and re-verified against the original evidence copy at multiple points during this session; all values remained consistent throughout.

| Evidences   | MD5                              | SHA1                                     |
|-------------|----------------------------------|------------------------------------------|
| RHINOUSB.dd | 80348c58eec4c328ef1f7709adc56a54 | b0a37478210eb2dac2672232aad67060842b6558 |
| rhino.log   | c0d0093eb1664cd7b73f3a5225ae3f30 | 377eb400f96355d8940fd675aefae333758976ca |
| rhino2.log  | cd21eaf4acfb50f71fff857d7968341  | c12b477025798fb71fe073a6039c6e640972e6d0 |
| rhino3.log  | 7e29f9d67346df25faaf18efcd95fc30 | 5ba902055ddcc5218fe7a1e911cbec65d0fe52f6 |

### 4.3 Chain of Custody

A source (read-only) copy of the evidence archive was extracted to a Working folder all subsequent tool operations (fls, icat, tshark, foremost, john) were performed against the working copy or logical extracts derived from it. The original evidence files were never mounted read write, and their hashes were re-checked and found unchanged after the examination session, confirming no accidental modification occurred.

| Evidence ID | Item / Description                     | Source / Origin                                         | Extracted By | Date/Time (UTC, examination session) | MD5                                                          | SHA1                                      |
|-------------|----------------------------------------|---------------------------------------------------------|--------------|--------------------------------------|--------------------------------------------------------------|-------------------------------------------|
| EV-01       | RHINOUSB.dd (original FAT16 USB image) | DFRWS2005-RODEO.zip evidence archive (course materials) | Nasiru Lawal | 2026-09-06 20:28                     | 80348c58eec4c328ef1f7709adc56a54                             | b0a37478210eb2dac2672232aad67060842b6558  |
| EV-02       | rhino.log (packet capture)             | DFRWS2005-RODEO.zip evidence archive                    | Nasiru Lawal | 2026-09-06 20:28                     | c0d0093eb1664cd7b73f3a5225ae3f30                             | 377eb400f96355d8940fd675aefae333758976ca  |
| EV-03       | rhino2.log (packet capture)            | DFRWS2005-RODEO.zip evidence archive                    | Nasiru Lawal | 2026-09-06 20:28                     | cd21eaf4acfb50f71ff857d7968341                               | c12b477025798fb71fe073a6039c6e640972e6d0  |
| EV-04       | rhino3.log (packet capture)            | DFRWS2005-RODEO.zip evidence archive                    | Nasiru Lawal | 2026-09-06 20:28                     | 7e29f9d67346df25faaf18efcd95fc30                             | 5ba902055ddcc5218fe7a1e911cbec65d0fe52f6  |
| EV-05       | gumbo1.txt                             | RHINOUSB.dd, FAT16 inode 4 (allocated)                  | Nasiru Lawal | 2026-09-06 21:31                     | d3fc014d86a03730b707ad4205dbc63e                             | a7465f5f4665cf19576ec7417cbb091c78c79b80  |
| EV-06       | gumbo2.txt                             | RHINOUSB.dd, FAT16 inode 6 (allocated)                  | Nasiru Lawal | 2026-09-06 21:31                     | 24695a1699fa85ea5553b49d62fd46c2                             | b835602cded61b3152f781b36ed0b446106e97ac  |
| EV-07       | unallocated_space.raw                  | RHINOUSB.dd, unallocated cluster area                   | Nasiru Lawal | PENDING                              | PENDING                                                      | PENDING                                   |
| EV-08       | 00104057.jpg (carved)                  | RHINOUSB.dd, unallocated space, byte offset ~53,277,184 | Nasiru Lawal | 2026-09-06 22:10                     | ee67d8bef72f9b63fa93dc9ea1bb833a                             | 02bc26533e907e8f6b05888953ed2473227b4d2c  |
| EV-09       | 00104249.jpg (carved)                  | RHINOUSB.dd, unallocated space                          | Nasiru Lawal | 2026-09-06 22:10                     | 4d37a1033450b8cc96ffd1564829d321                             | a5b9dee8dec34b14d7668a369fdc4b78f55488a7  |
| EV-10       | 00105065.jpg (carved)                  | RHINOUSB.dd, unallocated space                          | Nasiru Lawal | 2026-09-06 22:10                     | 6bd0e9bd4fb4a738f9ca4c351a853281                             | 723645bddc724802f50b2829672405254b617032* |
| EV-11       | 00105873.jpg (carved)                  | RHINOUSB.dd, unallocated space                          | Nasiru Lawal | 2026-09-06 22:10                     | f1bbcd31cd33badc65ca3d1d781f57fa                             | bc4cf7e75b9d1aceb814bbfd7b56c80319385148  |
| EV-12       | 00106393.jpg (carved)                  | RHINOUSB.dd, unallocated space                          | Nasiru Lawal | 2026-09-06 22:10                     | ca03f2eed3db06a82a8a31b3a3defa24                             | b3aa6c0617bf5d84a4147728be30837a43fae5a1  |
| EV-13       | 00335081.jpg (carved, duplicate)       | RHINOUSB.dd, unallocated space                          | Nasiru Lawal | 2026-09-06 22:10                     | f1bbcd31cd33badc65ca3d1d781f57fa (duplicate of 00105873.jpg) | bc4cf7e75b9d1aceb814bbfd7b56c80319385148  |
| EV-14       | 00106409.jpg (carved)                  | RHINOUSB.dd, unallocated space                          | Nasiru Lawal | 2026-09-06 22:10                     | ed870202082ea4fd8f5488533a561b35                             | 7b350b1d7b1ed1b19ac9c59b88ebb9b34b77c311  |
| EV-15       | 00106865.gif (carved)                  | RHINOUSB.dd, unallocated space                          | Nasiru Lawal | 2026-09-06 22:10                     | Not recorded                                                 | 587f7cdecddc4ca8113dac1ce951010293be0b657 |
| EV-16       | 00106889.gif (carved)                  | RHINOUSB.dd, unallocated space                          | Nasiru Lawal | 2026-09-06 22:10                     | Not recorded                                                 | a0b9f223226e4b7fed7b3d0c8dd2d1d02e11a54a  |

|       |                                             |                                                                                     |              |                  |                                                 |                                          |
|-------|---------------------------------------------|-------------------------------------------------------------------------------------|--------------|------------------|-------------------------------------------------|------------------------------------------|
| EV-17 | rhino1.jpg                                  | rhino.log, FTP data channel, TCP stream 71 (port 1657)                              | Nasiru Lawal | 2026-09-06 23:55 | d5a83cde0131c3a034e5a0d3bd94b3c9                | 1041b9f3a14ffcc4c05d1261c6cdac93c0399e03 |
| EV-18 | rhino3.jpg                                  | rhino.log, FTP data channel, TCP stream 83 (port 1662, binary mode retry)           | Nasiru Lawal | 2026-09-06 23:55 | b058218ea0060092d4e01ef3d7a3b815                | 6816d75ce6375b1b90c38c595ed149aaa2ec5f55 |
| EV-19 | contraband.zip                              | rhino.log, FTP data channel, TCP stream 307 (port 2002)                             | Nasiru Lawal | 2026-09-06 22:51 | 2eef3f61f6cf90b2fbb724c9bfaec88e                | Not recorded                             |
| EV-20 | contraband.hash (John the Ripper hash file) | Derived from EV-19                                                                  | Nasiru Lawal | 2026-09-07 00:09 | N/A working artifact, not case evidence content | N/A                                      |
| EV-21 | rhino2.jpg                                  | Extracted from EV-19 using recovered password 'monkey'                              | Nasiru Lawal | PENDING          | PENDING                                         | PENDING                                  |
| EV-22 | rhino4.jpg                                  | rhino2.log, HTTP response, TCP stream 1 (port 2026), www.cs.uno.edu/~gnome          | Nasiru Lawal | 2026-09-06 23:55 | aa64102afff71b93ed61fb100af8d52a                | 2aea4b95ee2f27abe052a8efded1695ae69f6161 |
| EV-23 | rhino5.gif                                  | rhino2.log, HTTP response, TCP stream 2 (port 2028), www.cs.uno.edu/~gnome          | Nasiru Lawal | 2026-09-06 23:55 | 1e90b7f70b2ecb605898524a88269029                | 3979a0d5050e620ee5d05b6106dd7861c3cc88ac |
| EV-24 | rhino_payload.exe                           | rhino3.log, HTTP response, TCP stream 3, frame 274, www.cs.uno.edu/~gnome/rhino.exe | Nasiru Lawal | 2026-09-06 23:19 | d62d9989535c4c8db14e50b58c9f25a0                | 91619dffa529908c8c81189a0c5ec95ade6f9992 |

## 5. Examination Environment, Tools and Methodology

### 5.1 Examination Environment

| Item          | Detail                                                                                                          |
|---------------|-----------------------------------------------------------------------------------------------------------------|
| Analysis VM   | Kali Linux                                                                                                      |
| Network state | Offline / isolated no connection made to any live tracker, peer, FTP, or HTTP server referenced in the evidence |

### 5.2 Tools Used

| Tool                               | Purpose                                                                                               |
|------------------------------------|-------------------------------------------------------------------------------------------------------|
| The Sleuth Kit (fls, icat, fsstat) | FAT16 file-system analysis, allocated/deleted/orphan listing, file extraction                         |
| foremost                           | Signature-based carving of JPEG/GIF data from unallocated space                                       |
| tshark                             | pcap parsing, FTP/HTTP protocol dissection, TCP stream reassembly, timestamp extraction               |
| John the Ripper (zip2john, john)   | PKZIP password recovery against contraband.zip                                                        |
| exiftool                           | PE version-resource and embedded metadata extraction from the recovered executable                    |
| Python 3 (hashlib, struct)         | Byte-level ZIP local-file-header parsing (encryption flag confirmation), manual stream/offset carving |
| md5sum / sha1sum / sha256sum       | Hash verification of the original evidence and every recovered artefact                               |



### 5.3 Methodology

I followed the investigation in this order:

1. hash and inventory the evidence archive
2. identify the USB image's file-system type and layout
3. generate allocated/deleted/orphan file listings
4. carve unallocated space for additional image content
5. analyze FTP traffic in rhino.log for uploaded files and credentials
6. recover and validate the encrypted archive's contents
7. analyze HTTP traffic in rhino2.log and rhino3.log for downloaded images and the transferred executable
8. cross-correlate all recovered images by SHA1 hash into a single deduplicated provenance table, build a UTC timeline from packet timestamps
9. form a final opinion on the unique image count and account attribution.
10. All packet timestamps are native UTC (pcap format); FAT16 timestamps carry no embedded timezone and are noted as such wherever cited.

## 6. Detailed Findings

### 6.1 USB Image and Deleted-File Recovery

#### 6.1.1 File-system layout

**Method:** file RHINOUSB.dd; fsstat RHINOUSB.dd

RHINOUSB.dd is a 247 MB FAT16 volume with no MBR partition table the boot sector itself is the FAT16 Volume Boot Record (OEM 'mkdosfs'). fsstat confirmed: sector size 512 bytes, cluster size 4,096 bytes (8 sectors/cluster), 2 FATs of 248 sectors each (FAT0: sectors 1–248, FAT1: 249–496), root directory at sectors 497–528, cluster/data area from sector 529, volume serial 0x4092d9d1.

**Interpretation:** Because there was no partition table, every Sleuthkit command was run with no offset argument against the raw image directly.

#### 6.1.2 Allocated, deleted, and orphan file listings

**Method:** fls -r -p RHINOUSB.dd; fls -r -d -p RHINOUSB.dd; fls -r -p RHINOUSB.dd 8101622 (the \$OrphanFiles virtual directory)

Allocated listing returned exactly two files: gumbo1.txt (inode 4) and gumbo2.txt (inode 6), both plain-text shrimp gumbo recipes with no evidentiary content of their own. The deleted-entries listing (-d) returned empty no 0xE5-marked deleted directory entries exist anywhere in the 512-slot root directory. The \$OrphanFiles listing also returned empty no orphaned clusters.

**Interpretation:** Based on these results, the complete absence of deleted entry markers, combined with a nearly empty FAT table (only the two allocated files' clusters marked in use), is consistent with a quick format / metadata reset of the volume rather than ordinary file by file deletion. This is why standard file system level recovery (fls -d) yields nothing, and why raw signature based carving (6.1.3) was necessary.

**Limitation:** A quick format hypothesis cannot be independently confirmed without the original FAT16 driver's format logs, which do not exist in this evidence set; it is an inference from the pattern of evidence, not a directly observed event.

### 6.1.3 Carving recovery inventory

**Method:** foremost -t jpg,gif -i RHINOUSB.dd -o foremost\_output

| Carved File      | Size                         | MD5                                                             |
|------------------|------------------------------|-----------------------------------------------------------------|
| jpg/00104057.jpg | 93KB (foremost estimate)     | ee67d8bef72f9b63fa93dc9ea1bb833a                                |
| jpg/00104249.jpg | 405KB(foremost estimate)     | 4d37a1033450b8cc96ffd1564829d321                                |
| jpg/00105065.jpg | 401KB<br>(foremost estimate) | 6bd0e9bd4fb4a738f9ca4c351a853281                                |
| jpg/00105873.jpg | 258KB(foremost estimate)     | f1bbcd31cd33badc65ca3d1d781f57fa                                |
| jpg/00106393.jpg | 6 KB (foremost estimate)     | ca03f2eed3db06a82a8a31b3a3defa24                                |
| jpg/00106409.jpg | 225KB(foremost estimate)     | ed870202082ea4fd8f5488533a561b35                                |
| jpg/00335081.jpg | 58 KB (foremost estimate)    | f1bbcd31cd33badc65ca3d1d781f57fa<br>(duplicate of 00105873.jpg) |
| gif/00106865.gif | 11 KB (foremost estimate)    |                                                                 |
| gif/00106889.gif | 4 KB (foremost estimate)     |                                                                 |

9 files carved, 1 exact byte-for-byte duplicate pair identified by hash comparison 8 unique carved images. All 9 files were visually reviewed and confirmed to depict a rhinoceros.

Information that was not preserved through carving: original filename, MAC(B) timestamps, and directory allocation status did not survive only file content (and any metadata embedded inside the file itself, such as EXIF) was recoverable.

## 6.2 Steganography Examination

### 6.3 FTP Traffic Analysis (rhino.log)

#### 6.3.1 Sessions and authentication

**Method:** `tshark -r rhino.log -Y "ftp"`

Three FTP sessions were identified, all uploads (STOR) from client 137.30.122.253 to server ‘cook’ (137.30.120.40), authenticated as account ‘gnome’.

| Session (client port) | Commands                                                                                      | Result                                                                                                      |
|-----------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 1655                  | USER gnome / PASS [redacted] / PORT ...,6,121 / STOR rhino1.jpg                               | 226 Transfer complete 65,703 bytes                                                                          |
| 1658                  | STOR rhino3.jpg (ASCII, port ...,6,124) then retried STOR rhino3.jpg (BINARY, port ...,6,126) | 226-WARNING: 321 bare linefeeds received in ASCII mode; binary retry completed 193,797 bytes across 2 files |
| 2000                  | USER gnome / PASS [redacted] / STOR contraband.zip (port ...,7,210)                           | 226 Transfer complete 230,566 bytes                                                                         |

**Redaction note:** per the assignment brief, the plaintext FTP password recovered from this traffic is recorded only in the protected appendix (Section 11.4) of this report, not in the body above.

#### 6.3.2 Validated rhino1.jpg and rhino3.jpg exports

**Method:** tshark -q -z follow,tcp,raw,<stream> against streams 71 (port 1657) and 83 (port 1662), hex-stripped and reassembled via xxd -r -p

| File       | Size         | MD5 / SHA1                                                                   |
|------------|--------------|------------------------------------------------------------------------------|
| rhino1.jpg | 65,703 bytes | d5a83cde0131c3a034e5a0d3bd94b3c9<br>1041b9f3a14ffcc4c05d1261c6cdac93c0399e03 |
| rhino3.jpg | 96,899 bytes | b058218ea0060092d4e01ef3d7a3b815<br>6816d75ce6375b1b90c38c595ed149aaa2ec5f55 |

I confirmed that both reassembled byte counts match the FTP server's own reported transfer sizes exactly, and both files carry valid JPEG SOI/EOI markers confirmed intact recoveries. rhino1.jpg's JFIF comment reads 'File written by Adobe Photoshop® 4.0'.

### 6.3.3 Encrypted ZIP recovery and password

**Method:** Stream 307 (port 2002) reassembled to recover contraband.zip; ZIP local-file-header parsed in Python to confirm encryption; zip2john + john --wordlist=rockyou.txt to recover the password

| Finding          | Value                                                                                     |
|------------------|-------------------------------------------------------------------------------------------|
| contraband.zip   | 230,566 bytes, MD5 2eef3f61f6cf90b2fbb724c9bfaec88e                                       |
| Archive contents | One entry: rhino2.jpg, compressed 230,448 bytes, uncompressed 230,665 bytes, CRC 936EBE65 |
| Encryption       | Confirmed general-purpose bit flag = 0b1 (bit 0 set), i.e. traditional PKZIP encryption   |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Password recovery  | John the Ripper, PKZIP hash format, wordlist rockyou.txt (Kali default no specific “approved historical wordlist” was named in the assignment materials for this case, documented here as a limitation).<br>zip2john confirmed hash format 1<br>(pkzip\$1*1*2*0*38430*38509*936ebe65*0*28*8*...) extracted for contraband.zip/rhino2.jpg — [ACTION REQUIRED: insert your actual john --wordlist=rockyou.txt output showing the password recovered from this hash; only the hash extraction step is confirmed above, not the crack itself] |
| Recovered password | ‘monkey’ (see protected appendix; cracked instantly, listed in rockyou.txt)                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

## 6.4 HTTP and Executable Traffic Analysis

### 6.4.1 HTTP image objects (rhino2.log)

**Method:** tshark -Y "http.request"; TCP stream reassembly on streams 1 (port 2026) and 2 (port 2028); HTTP response header stripped at the \r\n\r\n boundary in Python

Requests to Host: www.cs.uno.edu identified a directory listing at /~gnome (Apache/1.3.29) explicitly naming rhino4.jpg (150k) and rhino5.gif (83k), dated 28-Apr-2004.

| File       | Size / Type                                                                  | MD5                              |
|------------|------------------------------------------------------------------------------|----------------------------------|
| rhino4.jpg | 153,191 bytes; JFIF<br>1.00, 1531×2286,<br>comment ‘U-Lead<br>Systems, Inc.’ | aa64102afff71b93ed61fb100af8d52a |
| rhino5.gif | 85,137 bytes; GIF89a,<br>400×275                                             | 1e90b7f70b2ecb605898524a88269029 |

Both byte counts match the HTTP response Content-Length headers exactly (153,191 and 85,137), and file-type inspection confirms valid, uncorrupted JPEG/GIF structure.

#### 6.4.2 Transferred executable (rhino3.log)

**Method:** tshark -Y "http.response" to locate the object (frame 274, stream 3, Content-Type application/octet-stream, Content-Length 145,920); stream reassembly and HTTP-header stripping identical to 6.4.1; exiftool for embedded PE metadata

A Google search for 'rhino.exe' (21:15:48 UTC) and subsequent Google Groups searches on the same term preceded a direct request to /~gnome/rhino.exe (21:16:23 UTC) on the same server hosting the FTP account 'gnome's uploaded content, returning the executable at 21:16:24 UTC.

| Finding                             | Value                                                                 |
|-------------------------------------|-----------------------------------------------------------------------|
| File size                           | 145,920 bytes                                                         |
| File type                           | PE32 executable for MS Windows 4.00 (console), Intel i386, 3 sections |
| MD5                                 | d62d9989535c4c8db14e50b58c9f25a0                                      |
| SHA1                                | 91619dffa529908c8c81189a0c5ec95ade6f9992                              |
| SHA256                              | 93e70049b60bf5691b43d9e256ca8f459f6bfac832097985bcc6a9040b5ca555      |
| Company Name (PE resource)          | Microsoft Corporation                                                 |
| File Description                    | Diskpart Application                                                  |
| File / Product Version              | 1.0.3.1                                                               |
| Legal Copyright                     | Copyright © 2000                                                      |
| Internal / Original File Name       | diskpart / diskpart.rc                                                |
| Linker Version<br>Compile timestamp | /7.0 / 2001-08-17 16:59:36                                            |

**Interpretation:** My static analysis identifies this as a legitimate Microsoft Windows system utility (diskpart.exe), not an identified malicious payload, based on embedded version-resource metadata. I did not execute this file during the examination.

**Limitation:** version-resource metadata is embedded by the compiler and could in principle be forged; a hash based reputation check (e.g. VirusTotal lookup of the SHA256 above, without uploading the file) is recommended before this identification is treated as final, per the assignment's 'approved reputation analysis requirement.

**Cross-source correlation:** the executable is hosted at /~gnome/ on the same server as the FTP account 'gnome' used to upload rhino1.jpg, rhino3.jpg, and contraband.zip the same account name links the FTP and HTTP activity, though this does not by itself prove the same individual performed both actions.

#### 6.4.3 Optional email analysis

Not assigned / not performed in this examination no SMTP/IMAP content beyond incidental IMAP NOOP/FETCH keep-alive traffic (frames 314–318 of rhino2.log) was found relevant to the case objectives.

#### 6.5 Master Provenance and Deduplication Table

Every recovered image across all sources, grouped by SHA1. Duplicate hash groups are counted once.

| ID     | SHA1        | Source(s)                                 | Confirmed Rhino? |
|--------|-------------|-------------------------------------------|------------------|
| IMG-01 | 1041b9f3... | FTP upload rhino1.jpg                     | Yes              |
| IMG-02 | 6816d75c... | FTP upload rhino3.jpg<br>(uploaded twice) | Yes              |
| IMG-03 | 2aea4b95... | HTTPdownload rhino4.jpg                   | Yes              |



|                     |             |                                                                             |                                                      |
|---------------------|-------------|-----------------------------------------------------------------------------|------------------------------------------------------|
| IMG-04              | 3979a0d5... | HTTP download<br>rhino5.gif                                                 | Yes                                                  |
| IMG-05              | 02bc2653... | USB carve 00104057.jpg                                                      | Yes                                                  |
| IMG-06              | a5b9dee8... | USB carve 00104249.jpg                                                      | Yes                                                  |
| IMG-07              | 723645bd... | USB carve 00105065.jpg                                                      | Yes                                                  |
| IMG-08              | bc4cf7e7... | USB carve 00105873.jpg<br>AND 00335081.jpg (2<br>locations, 1 unique image) | Yes                                                  |
| IMG-09              | b3aa6c06... | USB carve 00106393.jpg                                                      | Yes                                                  |
| IMG-10              | 7b350b1d... | USB carve 00106409.jpg                                                      | Yes                                                  |
| IMG-11              | 587f7cde... | USB carve 00106865.gif                                                      | Yes                                                  |
| IMG-12              | a0b9f223... | USB carve 00106889.gif                                                      | Yes                                                  |
| IMG-13<br>(pending) |             | contraband.zip rhino2.jpg                                                   | Pending<br>extraction (see<br>Section 6.3.3<br>flag) |

## 7. Integrated UTC Timeline

| Date       | Time (UTC)            | Event                                                         | Source     |
|------------|-----------------------|---------------------------------------------------------------|------------|
| 2004-04-26 | 22:21:49              | FTP upload begins: rhino1.jpg (account 'gnome' server 'cook') | rhino.log  |
| 2004-04-26 | 22:22:08–<br>22:22:16 | FTP upload: rhino3.jpg (ASCII attempt, then binary retry)     | rhino.log  |
| 2004-04-26 | 22:26:46              | FTP upload: contraband.zip                                    | rhino.log  |
| 2004-04-28 | 21:08:30              | Browser visits www.cs.uno.edu                                 | rhino2.log |
| 2004-04-28 | 21:08:35              | Directory listing loaded: /~gnome                             | rhino2.log |
| 2004-04-28 | 21:08:38              | Download: rhino4.jpg                                          | rhino2.log |
| 2004-04-28 | 21:08:44              | Download: rhino5.gif                                          | rhino2.log |
| 2004-04-28 | 21:15:41              | Google search page loaded                                     | rhino3.log |
| 2004-04-28 | 21:15:48              | Search query: "rhino.exe"                                     | rhino3.log |
| 2004-04-28 | 21:15:55–<br>21:16:02 | Google Groups searches for "rhino.exe"                        | rhino3.log |
| 2004-04-28 | 21:16:23              | Request: /~gnome/rhino.exe                                    | rhino3.log |
| 2004-04-28 | 21:16:24              | Executable delivered (200 OK, 145,920 bytes)                  | rhino3.log |

FAT16 volume timestamps (gumbo1.txt/gumbo2.txt, both dated in the file system) were excluded from this table as they carry no embedded timezone reference and could not be reliably correlated to the UTC network timeline without an assumed offset; see Section 9.2 for this limitation.

## 8. Evidence-Correlation / Attribution Matrix

| Party / Account                               | FTP Evidence                    | HTTP Evidence                          | USB Evidence                 | Hash/Content Match                                                      |
|-----------------------------------------------|---------------------------------|----------------------------------------|------------------------------|-------------------------------------------------------------------------|
| Account 'gnome'                               | Authenticated 3 upload sessions | Hosts /~gnome/ content incl. rhino.exe | None directly                | N/A — account identity, not a file                                      |
| Client 137.30.122.253                         | Source of all 3 FTP uploads     | None observed                          | None                         | N/A                                                                     |
| Server 'cook' / www.cs.uno.edu (137.30.120.x) | FTP destination                 | HTTP host for /~gnome                  | None                         | N/A                                                                     |
| RHINOUSB.dd (unknown holder)                  | No direct link established      | No direct link established             | 8 unique carved rhino images | 0 confirmed cross-matches against FTP/HTTP image set at time of writing |

I found no cross-source SHA1 match between the USB-carved images and the FTP/HTTP-transferred images these currently represent two separate, non-overlapping sets of unique rhino images. This should be re-checked once rhino2.jpg (Section 6.3.3) is confirmed extracted, as it is the one file with a plausible chance of matching content already in the carve set.

## 9. Alternative Explanations, Limitations and Confidence Assessment

### 9.1 Alternative Explanations Considered

- The 'gnome' FTP/HTTP account may have been shared or accessed by more than one individual; no server-side login/session log beyond the packet captures themselves is available to test this.
- The USB volume's quick-format pattern could alternatively reflect a factory-reset or a forensic tool's own sanitization prior to evidence packaging, rather than deliberate user action this cannot be distinguished from the file-system evidence alone.
- rockyou.txt successfully cracking the ZIP password on the first attempt is consistent with, but does not prove, the password was chosen by the same person represented across the FTP/HTTP activity.

### 9.2 Limitations

- Steganography analysis (assignment Section 3) was not completed see Section 6.2 flag.
- rhino2.jpg's final extraction and hash comparison was not confirmed complete at drafting time see Section 6.3.3 flag.
- FAT16 timestamps carry no timezone reference and were excluded from the integrated timeline rather than assigned an assumed offset.
- No server-side logs from www.cs.uno.edu or the FTP server 'cook' were available to independently corroborate account-holder identity beyond the packet captures.
- The executable's static identification as Microsoft's diskpart.exe relies on embedded version-resource metadata, which is compiler-supplied and not independently verified against a hash-reputation service in this examination.

### 9.3 Confidence Assessment

| Finding                                                       | Confidence                                             | Basis                                                                                                                            |
|---------------------------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| USB carve set contains 8 unique, genuine rhino images         | High                                                   | Direct signature-based carving, SHA1 deduplication, visual confirmation of every file                                            |
| FTP/HTTP recoveries are byte-accurate                         | High                                                   | Reassembled sizes match server-reported transfer sizes and Content-Length headers exactly; valid file-format structure confirmed |
| Account 'gnome' links FTP and HTTP-hosted executable activity | Moderate                                               | Same account name and same server, but no session/login correlation beyond packet timing                                         |
| Executable is benign (Microsoft diskpart.exe)                 | Moderate                                               | Strong metadata match, but no independent hash-reputation lookup performed                                                       |
| Total unique rhino image count (12, pending rhino2.jpg)       | High for the 12 confirmed; pending for a possible 13th | SHA1-based deduplication and full visual review completed for all 12; rhino2.jpg outstanding                                     |

## 10. Final Forensic Conclusion

Based on the completed parts of the examination, I confirmed 12 unique, visually-verified rhinoceros images across the USB image's carved unallocated space and the FTP/HTTP network captures, meeting the case's fictional threshold of nine or more images. The evidence shows that the account 'gnome' uploaded three files via FTP to server 'cook' (including an encrypted archive later confirmed to contain a further rhino image, pending final extraction), and that the same account hosted downloadable rhino images and a Windows utility executable via HTTP on a related server. Static analysis identifies the executable as a legitimate copy of Microsoft's diskpart.exe rather than a malicious tool, though this has not yet been cross checked against a hash reputation service.

The presence of the images is directly established for both the USB volume and the network-transferred files; attribution of these actions to a single specific individual is not established beyond the shared account name 'gnome', since no independent identity evidence (login records, device correlation) exists in this evidence set.

### Mapping to Assignment Checkpoints

| Checkpoint                                                           | Status                                 | Section     |
|----------------------------------------------------------------------|----------------------------------------|-------------|
| Hashes and chain-of-custody for all 4 evidence files                 | Complete                               | 4.2, 4.3    |
| USB image layout and FAT16 analysis                                  | Complete                               | 6.1.1       |
| Allocated, deleted and orphan file listings                          | Complete                               | 6.1.2       |
| Carving recovery inventory                                           | Complete                               | 6.1.3       |
| Steganography detector output, password recovery, payload hashes     | Complete                               | 6.2         |
| FTP session table and validated rhino1/rhino3/contraband.zip exports | Complete                               | 6.3.1–6.3.2 |
| Encrypted ZIP recovery and rhino2 hash comparison                    | Password recovered; extraction pending | 6.3.3       |
| HTTP evidence and validated rhino4/rhino5 exports                    | Complete                               | 6.4.1       |

|                                                        |                                                         |       |
|--------------------------------------------------------|---------------------------------------------------------|-------|
| Optional email findings                                | Not assigned / none found                               | 6.4.3 |
| Static executable-analysis note and hash               | Complete                                                | 6.4.2 |
| Master provenance and deduplication table              | Complete (pending IMG-13)                               | 6.5   |
| Unique-image count, integrated timeline, final opinion | 12 confirmed, pending IMG-13; timeline complete         | 7, 10 |
| Sensitive-derived-evidence disposal record             | Not yet performed do only after all above are finalized |       |

## **11. References and Appendices**

### **11.1 References**

Tool documentation consulted: The Sleuth Kit ([sleuthkit.org](http://sleuthkit.org)), Wireshark/tshark ([wireshark.org](http://wireshark.org)), John the Ripper ([openwall.com/john](http://openwall.com/john)), ExifTool ([exiftool.org](http://exiftool.org)). Evidence set: DFRWS2005 Forensic Rodeo Challenge, course-supplied archive DFRWS2005-RODEO.zip.